

Statistical Programming Languages

Reading and Writing Data

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Data management

- ▶ Sources of data
 - ▶ Data in human readable format (.csv, .txt)
 - ▶ Data in binary format (Excel, SPSS, STATA, etc)
 - ▶ Data from relational databases (Oracle, SAP, etc)
- ▶ R has more than 100 built-in datasets, which can be displayed by `ls("package:datasets")`

```
head(ls("package:datasets"))  
## [1] "ability.cov"    "airmiles"  
## [3] "AirPassengers" "airquality"  
## [5] "anscombe"      "attenu"
```

- ▶ Many packages provide their own datasets
- ▶ `data()` shows all datasets of all loaded packages

Loading R-provided data

```
data(iris)      # load dataset iris; see ?iris
head(iris, 4)   # object containing the iris data
##   Sepal.Length Sepal.Width Petal.Length
## 1           5.1           3.5           1.4
## 2           4.9           3.0           1.4
## 3           4.7           3.2           1.3
## 4           4.6           3.1           1.5
##   Petal.Width Species
## 1          0.2  setosa
## 2          0.2  setosa
## 3          0.2  setosa
## 4          0.2  setosa
class(iris)
## [1] "data.frame"
```

Handling external Data

- ▶ Two main functions:
 - ▶ `read.table()` and `write.table()`: very general, for all kinds of text-based data
 - ▶ `read.csv()` and `write.csv()`: wrappers for `*.table()` for .csv files
- ▶ Connection to Excel, SQL, and other data sources through specific packages

Reading data: `read.table()`

```
read.table(file, header, sep, dec, row.names, ...)
```

- ▶ `file`: the name of the file to be read (including file extension)
- ▶ `header`: a logical value indicating whether the file contains the names of the variables as its first line
- ▶ `sep`: the field separator character (tabulated, comma, semicolon, space, etc.)
- ▶ `dec`: the character used in the file for decimal points
- ▶ `row.names`: a vector of row names

Similarly with `read.csv`.

Writing data: `write.table()`

```
write.table(x, file, sep, na, dec, row.names, ...)
```

- ▶ `x`: the object to be written, preferably a matrix or data frame
- ▶ `file`: a character string naming the file (including file extension)
- ▶ `sep`: the field separator string
- ▶ `na`: the string to use for missing values in the data
- ▶ `dec`: the (single) character to use in the file for decimal points
- ▶ `row.names`: logical variable. Default TRUE

Similarly with `write.csv`

Accessing sub- and superordinated folders

- ▶ Per default R accesses your working directory to read or write data
- ▶ You can access a file in a subfolder by `file =`
`"./Subfolder/my_file.*"`
- ▶ You can access a file in the parent folder by `file =`
`"../my_file.*"`
- ▶ You can also use the respective data path to access a file
(components of a path have to be separated by `/`)

```
diris <- read.table(file = "iris.txt", header = TRUE,
                    sep = " ", dec = ".",
                    row.names = NULL)

class(diris)
## [1] "data.frame"
head(diris, 2)
##   Sepal.Length Sepal.Width Petal.Length
## 1          5.1          3.5          1.4
## 2          4.9          3.0          1.4
##   Petal.Width Species
## 1          0.2  setosa
## 2          0.2  setosa
dim(diris)
## [1] 150  5
new.diris <- diris[diris[, "Sepal.Width"] >= 3, ]
dim(new.diris)
## [1] 93  5
write.table(new.diris, file = "niris.txt", sep = " ",
            na = "NA", dec = ".", row.names = FALSE)
```


summary() function

```
summary(new.diris)
```

```
##      Sepal.Length      Sepal.Width      Petal.Length
##  Min.      :4.300    Min.      :3.000    Min.      :1.000
##  1st Qu.:5.000    1st Qu.:3.000    1st Qu.:1.500
##  Median :5.600    Median :3.200    Median :1.900
##  Mean   :5.776    Mean   :3.313    Mean   :3.298
##  3rd Qu.:6.500    3rd Qu.:3.500    3rd Qu.:5.200
##  Max.   :7.900    Max.   :4.400    Max.   :6.700
##      Petal.Width      Species
##  Min.      :0.100    Length:93
##  1st Qu.:0.200    Class :character
##  Median :0.500    Mode  :character
##  Mean   :1.046
##  3rd Qu.:1.800
##  Max.   :2.500
```

Reading files in R from other sources

- ▶ Excel files: `read.xlsx2()` from the package `xlsx` or `read_excel()` from the package `readxl`
- ▶ SPSS files: `read.spss()` from the package `foreign`
- ▶ SAS files: `read.sas7bdat()` from the package `sas7bdat`
- ▶ EViews files: `readEViews()` from the package `hexView`
- ▶ etc.